JAL PATEL

Calgary, AB | (587) 839-4452 | jal.patel2@ucalgary.ca | linkedin.com/in/jal-patel123

EDUCATION

University of Calgary Calgary, AB, Canada

Bachelor of Science in Mechanical Engineering *Expected April 2028*

SKILLS

CAD & Design: SolidWorks, AutoCAD, GD&T, tolerance analysis, structural FEA

Programming & Modeling: Python, MATLAB/Simulink, Simscape Battery, equivalent-circuit modeling, Excel

Battery & Testing: HPPC testing, cell parameter extraction, pack simulation, torque/speed/temperature measurement

Manufacturing: Manual lathe, manual mill, waterjet, welding, 3D printing, assembly

EXPERIENCE

Undergraduate Research Assistant – Robotic Joint Manufacturing & Servo Control *Sep 2026 – Present*

University of Calgary

- **Integrated** a **harmonic-drive reducer**, **Mitsubishi servo motor**, and custom mechanical components to develop a **PLC-controlled robotic joint** for **current and vibration diagnostics**.
- Wire & configure a **Panasonic FP-X0 PLC, MR-JE-40A servo amplifier,** and limit switches to achieve motor rotation control.
- **Developed and tested** ladder-logic programs in **Control FPWIN Pro7** to implement **forward/reverse motion** and establish control sequences for current and vibration diagnostics.

Electric Drivetrain & Powertrain Engineering Lead *Sep 2024 – Present*

Zeus Electric Motorsport

- **Lead and mentor a 10+ member powertrain team** across mechanical design, CAD, manufacturing, analysis, fabrication, and testing; coordinate design reviews, assign technical tasks, and support engineering decisions throughout the development process.
- Designed and manufactured a custom drivetrain sprocket and motor dynamometer fixture in **SolidWorks** to support dyno testing up to **170 Nm** peak torque, with FEA-validated sprocket safety factor **>2.0** under peak load.
- **Coordinate** with an **11-member** battery sub-team to design and fabricate a liquid-cooling loop for a **96 V NMC battery pack** comprising **30 cell modules**, targeting cell temperatures below **35 °C** and pack temperature variation under **5 °C** during high-load operation.

Research Intern – Electrochemical Diagnostics & Cell Modeling *May 2026 – Aug 2026*

Technische Universität Berlin

- Characterized **30 lithium-ion cells** using **HPPC testing** and automated **MATLAB** pulse detection, processing approximately **100 hours** of cycler data per cell to extract consistent OCV, resistance, and time-constant parameters.
- Built a **6-stage MATLAB pipeline** to extract, estimate, validate, create, and package individual cell parameters, enabling automated conversion of raw cycler files into a fully parameterized pack simulation.
- Developed a single-cell **Simulink** model with **CC-CV cycling** and an **adaptive Kalman filter** to estimate state of charge from noisy sensor measurements, replicating realistic charge discharge protocols and BMS style estimation.
- Fit individual **1-RC equivalent-circuit models** for all **30 cells** using cell-specific HPPC data, achieving typical validation errors of **15.2 mV at 1C** and **24.4 mV at 2C** over the **10–90% SOC** operating range

Formula Manufacturing & Machining Engineering Student *May 2026 – Aug 2026*

FASTTUBE

- **Interpreted GD&T callouts**, ANSI/ISO standards, and engineering drawings to develop manufacturing plans for custom components while meeting specified dimensional tolerances.
- **Machined custom and competition-critical replacement components** from raw stock using **manual lathes and mills**, completing parts under tight deadlines and drawing requirements.
- **Prioritized competition-critical jobs** and organized part inventory and tooling to ensure materials and equipment were available for scheduled fabrication work.

TECHNICAL PROJECTS

Autonomous Payload Retrieval Robot | Team Lead *Jan 2025 – May 2025*

- Led mechanical design and system integration of an autonomous robot, developing **CAD models**, wiring diagrams, and MATLAB/Python simulations that improved route completion performance by **25%**.

Smart Hydroponic Cultivation System | Mechatronic Systems Member *Jan 2025 – May 2025*

- Built an automated hydroponic system with moisture, pump, and lighting sensors, using **MATLAB/Excel control optimization** to improve energy efficiency by **15%** and designing a **3D-printed enclosure** for high-humidity operation.

LEADERSHIP & EVENT MANAGEMENT

Engineering Student Society | Competition Commissioner *May 2025 – June 2026*

- Orchestrated engineering competitions for **115+ participants** by coordinating **52 volunteers**, securing **$10,000 in sponsorships**, managing a **$9,750 budget** with less than **2% variance**, and increasing registrations **42%** year over year